

PHINMA UNIVERSITY OF ILOILO
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Xtrip: Laser Tripwire Alarm System

A Capstone Project

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College of Information Technology Education

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In Partial Fulfillment

Of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled "**Laser Tripwire Alarm System**" prepared and Submitted by **Luzel A. Ambugana, Julian Jacob P. Casimiro, and Dan Rytchy Magneser** who are hereby recommended for corresponding final evaluation.

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Acknowledgment

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Abstract

As we leave our residences or stores, we can't guarantee that it is safe from criminals. An unauthorized entry to our place can lead to a potential risk of our place being accessed by criminals.

Improving our security is the best way for us to get protected. Therefore, we are creating a tripwire that detects unauthorized entry to our places. This is the "Laser Tripwire Alarm System". It is an Arduino-based project that uses a laser module and sensor. The laser beam is projected to the sensor and can be triggered when an intruder passes through it, resulting in an alarm. Another feature is a web application where you can navigate the alarm system. You can choose whether you can turn the alarm armor on or off.

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CHAPTER 1

INTRODUCTION

The goal of this capstone project is to design and implement the Xtrip Laser Tripwire Alarm System, a robust intrusion detection solution that provides real-time monitoring and alerts for enhanced security in residential and commercial environments. This system addresses the growing need for affordable security measures by overcoming the limitations of traditional solutions, such as high costs, installation complexity, and the lack of immediate visual evidence during breaches. By integrating an ESP32 microcontroller with an ESP32-CAM module, the Xtrip system enables precise laser-based tripwire detection, instant auditory alarms via a buzzer, and automated image or video capture to document intrusions, ultimately strengthening the protection of sensitive areas like stock rooms in businesses and resorts.

The project emphasizes a user-friendly setup that combines low-cost hardware with IoT capabilities for seamless operation. The ESP32 serves as the central processing unit, handling sensor data from a laser module and light-dependent resistor to detect beam interruptions, while the ESP32-CAM facilitates wireless video streaming and evidence storage on an SD card or cloud platform.

In addition, the Xtrip system incorporates advanced elements like object detection using models ESPE2 CAM for identifying intruders via the camera feed, enhancing accuracy over basic tripwire setups. This adaptability makes it suitable for diverse applications, from small retail spaces to large resort properties, while prioritizing energy efficiency and minimal wiring.

This project, the Xtrip Laser Tripwire Alarm System, will empower users to achieve proactive security management, reducing response times to threats, minimizing false alarms through reliable laser detection, and providing evidentiary support for investigations, thereby boosting overall safety and operational peace of mind.

TECHNICAL BACKGROUND OF THE STUDY

The Xtrip system is founded on the established principle of laser tripwire technology, where an unbroken laser beam forms an invisible security perimeter. Any interruption of this beam signifies a potential breach, triggering predefined security protocols. Essential components of such systems typically include a laser emitter, a light-sensitive sensor and a processing unit responsible for interpreting sensor data and initiating responses.

The ESP32 microcontroller serves as the central intelligence of the Xtrip system, lauded for its potent processing capabilities, integrated Wi-Fi, and Bluetooth connectivity, which are indispensable for modern Internet of Things (IoT) applications requiring remote monitoring and instant alerts. Furthermore, the integration of the ESP32-CAM module provides an embedded camera functionality, enabling the system to capture images upon detecting an intrusion, thereby furnishing invaluable visual evidence. This integration of laser detection with an intelligent, connected camera system positions Xtrip as a comprehensive and advanced security solution.

STATEMENT OF THE PROBLEM

Businesses, particularly resorts with numerous restricted areas like stock rooms, face significant challenges in preventing unauthorized access. Current security measures often suffer from several shortcomings:

- Delayed Detection and Response: Traditional systems might not provide immediate alerts, leading to potential loss or damage before intervention.
- Lack of Visual Evidence: Many basic alarm systems only trigger an audible alert, failing to provide photographic or video evidence of the intruder, making identification and prosecution difficult.
- High Costs: Implementing and maintaining advanced security systems can be prohibitively expensive for some businesses.
- False Alarms: Systems prone to false alarms can lead to alarm fatigue and wasted resources.
- Complexity in Integration: Integrating disparate security components into a unified, remote-monitorable system can be complex.

The Xtrip Laser Tripwire Alarm System aims to solve these problems by offering an affordable, reliable, and integrated solution that provides real-time intrusion detection, immediate notification to property owners, and crucial visual evidence capture.

OBJECTIVES OF THE STUDY

- To design the hardware architecture of the Xtrip system, incorporating a laser emitter, a light-sensitive sensor, an ESP32 microcontroller, an ESP32-CAM module, and an alarm buzzer.
- To develop the firmware for the ESP32 that accurately detects laser beam interruptions, triggers an audible alarm, and controls the ESP32-CAM for image capture.
- To implement a secure and reliable real-time notification mechanism to alert property owners of intrusions, potentially including captured images.
- To evaluate the Xtrip system's performance metrics, including detection accuracy, response time, false alarm rate, and reliability in various simulated environments.
- To gather and analyze user feedback on the system's usability, effectiveness, and overall satisfaction.

SIGNIFICANCE OF THE STUDY

This study is significant because it addresses common challenges in securing restricted areas, where traditional security measures like locks and basic cameras often fall short in providing real-time detection and evidence of intrusions. Manual monitoring or expensive commercial systems can lead to delayed responses, higher costs, and vulnerabilities in sensitive environments such as business premises, resorts, or private properties. By developing the Xtrip Laser Tripwire Alarm System, this study aims to improve the reliability, affordability, and intelligence of intrusion

detection using IoT-enabled components like ESP32 and ESP32-CAM, making advanced security accessible for various applications in restricted areas.

The key benefits of this study include:

- **Enhanced Protection for Restricted Areas:** The Xtrip system delivers a cost-effective and highly effective tool for safeguarding sensitive zones, such as storage vaults or private lounges, by reducing inventory loss and preventing unauthorized access. Its ESP32-CAM integration captures visual evidence for thorough incident investigations, outperforming basic alarms by providing actionable proof.
- **Immediate Alerts and Response for Property Owners:** Owners of resort or estates benefit from peace of mind with instant Security when the laser beam is interrupted, allowing quick intervention to minimize theft or damage. This real-time visual confirmation via the camera module ensures breaches are documented without constant human oversight.
- **Innovation in IoT and Embedded Systems for Security:** The project illustrates a practical use of ESP32 and ESP32-CAM to build connected, smart solutions that detect beam interruptions and trigger alerts, advancing the field by showing how low-cost microcontrollers enable scalable IoT security networks. This contributes to broader IoT adoption in intrusion detection, integrating sensors like LDR for precise monitoring.
- **Deterrence and Prevention of Intrusions:** The visible or invisible laser setup acts as a psychological deterrent, while automated buzzer or LED alerts scare off potential intruders in real-time, reducing the likelihood of breaches in high-risk zones like industrial sites.

- **Improved Data Integrity and Legal Compliance:** By capturing timestamped photos or videos upon detection, the system aids in forensic analysis and legal proceedings, ensuring tamper-evident records that comply with security standards in regulated industries.
- **Restricted areas:** Provides a cost-effective and highly effective tool for securing sensitive areas, reducing inventory loss, and preventing unauthorized access. The visual evidence capability assists in incident investigation.

Scope and Limitation of the Study

This study focuses on the development of the Xtrip Laser Tripwire Alarm System, an innovative IoT-based intrusion detection prototype designed to enhance security for restricted areas, such as resort stock rooms. The system employs laser tripwire technology to create an invisible barrier, triggering alerts upon beam interruption. Key features include real-time detection using a laser emitter and light-dependent resistor (LDR) sensor, integrated with an ESP32 microcontroller for processing and IoT connectivity, and an ESP32-CAM module for capturing visual evidence.

including:

- **Laser Tripwire Detection:** Establishing a continuous low power laser beam across short-to-medium range perimeters (up to 10-20 meters indoors), with an LDR sensor to detect interruptions and initiate responses.
- **Microcontroller Integration:** Utilizing the ESP32 for data processing, Wi-Fi/Bluetooth communication, and control of peripherals like the buzzer for audible alarms.

- **Visual Evidence Capture:** Employing the ESP32-CAM to take still images or short video clips (up to 15 seconds) upon detection, stored locally or transmitted via IoT.
- **Real-Time Notifications:** Implementing IoT protocols to send alerts via email or mobile app or similar platforms to property owners, enabling remote monitoring and rapid response.

The system is tailored for indoor settings like businesses, resorts, or small facilities, emphasizing cost-effectiveness and ease of deployment. The scope includes hardware assembly, software programming, testing for reliability in controlled conditions, and assessment of its impact on reducing unauthorized access and inventory loss.

Limitations of the Study

While the study aims to deliver a functional and efficient laser tripwire prototype, several limitations constrain its scope and performance:

- The current system may not incorporate advanced intruder recognition (facial recognition, gait analysis). Image capture is limited to still photos or short clips and not continuous high-definition video streaming, due to potential storage and bandwidth constraints.
- Performance may be affected by extreme environmental factors such as dense fog, heavy rain, or direct, intense sunlight potentially interfering with the laser beam or sensor, if not adequately shielded.
- The power supply typically requires a stable AC source or a reliable battery system with limited prolonged off-grid operation without significant optimization.

- The range and reliability of the laser detection are dependent on the specific laser module power and the sensitivity of the chosen light sensor.

Definition of Terms

- Xtrip - The developed prototype of the Laser Tripwire Alarm System, an IoT-enabled security solution using ESP32 hardware for intrusion detection in restricted areas.
- Laser Tripwire - A security mechanism that projects a continuous laser beam to form an invisible barrier; interruption of the beam by an object or person triggers detection and alerts.
- Buzzer - An audible alerting device that produces sound signals upon intrusion detection, serving as an immediate on-site warning to deter intruders or alert nearby personnel.
- ESP32 - A powerful, low-cost System-on-Chip (SoC) microcontroller with integrated Wi-Fi and Bluetooth, commonly used in IoT projects.
- ESP32-CAM: A small-sized ESP32-based development board with an integrated camera module, capable of image capture and streaming.
- Intrusion Detection - The process of identifying unauthorized entry or activity within a protected area.
- IoT (Internet of Things) - A network of interconnected physical devices embedded with sensors, software, and other technologies, enabling them to connect and exchange data over the internet.
- LDR (Light Dependent Resistor) - A passive component

whose resistance changes based on the intensity of light falling on it, often used as a light sensor in tripwire systems.

- Firmware - Specialized software embedded in a hardware device to control its specific functions.
- Perimeter Security - Protective measures focused on the boundaries of a defined area, such as laser barriers, to prevent unauthorized entry into spaces like stock rooms or facilities.
- Beam Interruption - The event where an object crosses the laser path, reducing light intensity on the LDR sensor and activating the system's response mechanisms.
- False Positive - An erroneous alarm triggered by non-threat factors, such as environmental interference, highlighting a limitation in basic sensor-based detection without advanced filtering.
- Restricted Areas - Designated zones requiring heightened security, such as resort stock rooms, where the system prevents theft or unauthorized access to valuable inventory.

CHAPTER II

Review of Related Literature

Overview

As the need for effective and real-time monitoring grows in both residential and commercial settings, security systems continue to change. Automated solutions that incorporate sensors, cameras, and web-based technologies are gradually replacing manual observation-based traditional systems.

Devices can now communicate and share data over networks thanks to the development of the Internet of Things (IoT), a concept first proposed by Kevin Ashton. By recording events and sending data to centralized platforms for monitoring and analysis, this development enables security systems to function intelligently.

The suggested system offers a cutting-edge method of intrusion detection by combining a laser tripwire mechanism with image capture and database logging. The system guarantees effective security event monitoring and documentation through the integration of web technologies and embedded systems.

Traditional VS Modern Styles

Traditional security systems mainly rely on manual procedures and simple electronic parts like standalone CCTV cameras and alarms. These systems don't offer real-time remote access to data and frequently need human supervision.

However, to improve functionality, modern systems make use of automation, sensors, and internet-based technologies. These systems have the ability to take pictures, identify intrusions, and store information in databases that web apps can access from a distance.

Technology Use

The system integrates multiple technologies to achieve its functionality:

Microcontroller Unit

The ESP32 serves as the core processing unit of the system. It handles input from the laser sensor, processes detection logic, and communicates with the web server via Wi-Fi. Its built-in networking capability makes it suitable for IoT-based applications.

Camera Integration

The ESP32-CAM is used to capture images whenever the laser beam is interrupted. This ensures that every intrusion event is documented visually, improving the reliability of the system as evidence can be reviewed later.

Laser Tripwire Mechanism

The laser tripwire system consists of a laser emitter and a receiver (e.g., LDR or photodiode). When the beam is interrupted, the system detects a change in signal and triggers the image capture process. This method provides precise and immediate detection of intrusions.

Web-Based Monitoring System

A web application is used to display and manage recorded data. The system logs key information such as time, date, intrusion status, and captured images into a database. Users can access this information remotely, enabling efficient monitoring without physical presence.

Database System

The database stores all captured events in a structured format. Each record includes fields such as timestamp, detection status, and image data. This ensures proper documentation, easy retrieval, and analysis of intrusion events.

CASE STUDIES

The following studies support the design and implementation of the proposed system:

Case Study 1: IoT-Based Surveillance Systems

A study by Rajkumar Buyya demonstrated that IoT-enabled surveillance systems improve monitoring efficiency by integrating sensors, cameras, and web platforms. The research emphasized real-time data access and automation.

Case Study 2: Embedded Camera Security Systems

Research published in the International Journal of Computer Applications showed that embedded systems with camera integration enhance security by providing visual evidence during intrusion events. This aligns with the image capture feature of the proposed system.

Case Study 3: Laser-Based Intrusion Detection Systems

A study from the Institute of Electrical and Electronics Engineers explored laser-based detection systems and found them to be highly accurate for perimeter security. The system's ability to detect beam interruption with minimal delay supports its use in real-time applications.

Case Study 4: Web-Based Monitoring Systems

According to Andrew S. Tanenbaum, network-based systems enable efficient remote monitoring and data management. Web-based platforms allow users to access system data anytime, which supports the proposed system's remote monitoring capability.

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CHAPTER III

Methodology

This chapter outlines the research methodology that will be employed in the development of the Xtrip Laser Tripwire Alarm System. The primary goal of this research is to design and implement an advanced, automated intrusion detection system that integrates technologies such as the Internet of Things (IoT) embedded microcontrollers and real-time notification mechanisms. This chapter will detail the research approach, data collection methods, system design, tools, and techniques that will be used to achieve this objective.

The research will adopt a systematic approach to provide the effective development and implementation of the proposed system thorough review of existing intrusion detection systems will be conducted to identify gaps and areas for improvement. Based on this review, the methodology will proceed with the design phase, where the architecture and components of the Xtrip system will be defined.

To gather valuable insights on user requirements and expectations, surveys and interviews will be conducted with key respondents, including industry professionals and experts in the field of security management. These respondents will play a crucial role in providing the necessary data for the system's design and functionality. Their feedback will guide the development process, ensuring the system meets the practical needs of users.

Once the system is developed, it will undergo rigorous testing to assess its performance, usability, and efficiency. This chapter will provide a clear understanding of the research methods and processes that will guide the study, ensuring that the project is executed in a structured and effective manner, leading to meaningful results in the field of IoT-based security systems.

Research and Development Stage

Understanding the current security landscape, user needs and existing technologies formed the bedrock of the Xtrip Laser Tripwire Alarm System Research and Development (R&D) stage. We initiated this phase by thoroughly reviewing the strengths and weaknesses of various intrusion detection technologies focusing on laser-based systems, passive infrared (PIR) sensors and camera-based surveillance. Laser tripwire technology was identified for its ability to create an invisible, precise, and easily definable perimeter, making it highly suitable for specific access control applications such as restricted areas.

Following this initial research, the design phase concentrated on integrating the core laser detection mechanism with advanced monitoring and evidence collection capabilities.

We studied how microcontrollers like the ESP32 and ESP32-CAM are implemented in real-world IoT and security applications to guide our system's architecture. The ESP32 was selected for its robust processing power and integrated Wi-Fi connectivity, essential for remote monitoring and instant

alerts while the ESP32-CAM was chosen for its embedded camera functionality, crucial for capturing visual evidence of intrusions Using

ESP32CAM. Simultaneously, we researched various notification methods, including email, SMS, and dedicated mobile app integrations, to ensure timely and effective communication with property owners. This step was crucial in ensuring that alerts are not only immediate but also accompanied by verifiable visual context.

The development stage commenced with building the laser detection module focusing on optimizing the alignment between the laser emitter and the light-sensitive receiver (LDR) Light Dependent Resistors or photodiode and refining the sensitivity threshold in the ESP32 firmware. We evaluated different light sensing techniques to select a method that was both reliable and less prone to false alarms caused by environmental factors. After establishing stable detection, we implemented the camera capture and image transmission modules, ensuring that images were taken promptly upon detection and securely transmitted. These components were rigorously tested to prevent any potential missed captures or delays in notification delivery.

Remote monitoring and notification capabilities were developed next focusing on ensuring that alerts including captured images could be securely and reliably sent to the intended recipient. We incorporated Wi-Fi connectivity to integrate with existing network infrastructures, allowing the system to communicate with a designated mobile application or email service. User interface elements for arming/disarming and managing settings were also developed,

Throughout the R&D process we conducted continuous testing including functional testing like verifying detection accuracy under various conditions. reliability testing like prolonged operation without failure and performance benchmarking measuring response times. Feedback from security managers and experts gathered through surveys and interviews, was continuously incorporated to refine the system's design, enhance its usability and ensure it met the practical requirements of real-world security management. The final Xtrip Laser Tripwire Alarm System reflects a balance between precise intrusion detection immediate notification and verifiable visual evidence ensuring it effectively meets the needs of businesses requiring secure monitoring of restricted areas.

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a structured approach used in the development of the "Xtrip Laser Tripwire Alarm System" to develop a project that is completed systematically, efficiently, and effectively. The SDLC will guide the development process from the initial planning phase to the final deployment, ensuring that the system meets the needs of the users and aligns with the insights provided by the respondents.

Key Features of Iterative Prototyping:

- **Iterative Development:** The system was developed in small, manageable increments called prototypes, allowing for

progressive enhancement of features like laser detection and image capture.

- **Continuous Feedback:** Regular feedback from stakeholders was incorporated to refine system features, such as optimizing sensor thresholds based on real world testing.
- **Collaboration:** Close collaboration between developers, users, and project advisors ensured alignment with user needs, particularly for security managers in restricted environments.
- **Flexibility:** prototyping allowed for changes in requirements even during the later stages of development, accommodating hardware adjustments or new IoT integrations.

The SDLC will consist of the following stages:

- **Planning and Requirement Analysis**

The first phase of the SDLC involves gathering detailed requirements for the system. In this stage, the input from the respondents, including security managers and industry experts, will be essential in identifying the specific challenges and needs that the system must address. Through surveys, interviews, and discussions, the researchers will gather data on key features, desired functionalities, and the technical requirements of the laser tripwire alarm system. This phase will also involve setting clear goals, defining the project scope, and developing a timeline for the project.

- **System Design**

In the design phase, the systems architecture and overall structure will be defined. The respondents feedback will guide the design of both the hardware components such like

laser sensor integration and firmware logic intrusion detection algorithms. The design will also incorporate modern technologies such as IoT for remote alerts, ESP32 for processing, and ESP32CAM for visual capture, ensuring that the system is both scalable and user-friendly. The design phase will focus on ensuring that the system meets the functional and non-functional requirements identified in the previous phase.

- **Development**

During the development phase, the actual coding and implementation of the system will take place. Based on the designs, the developers will create the firmware using appropriate programming languages and tools. The respondents will continue to play an important role by reviewing early prototypes and providing feedback on features such as usability, accuracy, and performance. This iterative process will help the system be aligned with user needs and expectations.

- **Testing**

Once the system is developed, Testing will involve both functional testing to check if all features are working correctly and performance testing to test if the system can handle various environmental conditions and usage. The respondents will be involved in user acceptance testing (UAT) User Acceptance Testing where they will test the system in real-world conditions to identify any issues or areas for improvement. Feedback from this phase will help fine-tune the system before its full deployment.

- **Deployment**

After successful testing, the system will be deployed in a live environment. This phase involves installing and

configuring the system for actual use, ensuring that it is fully operational and accessible to users. The deployment phase will also involve providing initial training to users, so they can effectively operate the system. Respondents will be consulted during the deployment process.

- **Maintenance and Support**

Respondents will provide feedback on system performance and usability, helping to identify areas for future updates or improvements. Regular updates, bug fixes, and additional features will be integrated based on the evolving needs of users and the industry.

Data Collection Techniques

In order to gather the necessary insights for developing the "Xtrip Laser Tripwire Alarm System," a variety of data collection techniques will be employed. These techniques will help the system meet the needs of users, particularly security managers and other key stakeholders, while also addressing current challenges in intrusion detection. The following data collection methods will be used:

- **Literature Review**

Literature review will be used as a primary method for collecting foundational knowledge from a wide range of sources, including security experts, academic papers, and technical documentation. The review will be designed to capture information on current intrusion detection practices, challenges, and expectations for an improved system. The sources will be structured to gather both theoretical insights and practical details on key features, system requirements, and performance expectations. This will provide valuable insights into the technologies that should

be prioritized in the Xtrip laser tripwire alarm system.

- Experimental Testing

Hands-on experimental testing will be conducted with selected components to collect quantitative data. These tests will be chosen based on their relevance to security applications, focusing on hardware like laser modules and light sensors. The experiments will allow for a deeper understanding of challenges faced by security systems, their operational workflow, and how an advanced tripwire system could improve detection accuracy and response time. During these tests, controlled variables will encourage detailed measurements, offering insights that may not be captured through theoretical review alone.

- Prototyping Logs

Prototyping logs will be organized to document iterative builds and gather a variety of perspectives on the system's evolution. These logs will enable the research team to record hardware assembly, code versions, and test outcomes in a collaborative manner. Prototyping logs will allow for interaction between development phases, providing a platform to share observations, modifications, and results in a structured environment. The documentation from these logs will be valuable for refining system components and ensuring they align with user needs.

- Surveys

Surveys will be conducted to understand the operational aspects of security management in real-world settings. This technique involves querying security personnel at different organizations to identify vulnerabilities, pain points, and areas that could benefit from automation or real-time

alerts. By surveying practices directly, the research team will gain a better understanding of how tripwire systems are used in practice and what improvements can be made.

- Document Analysis

To further inform the system design, existing documents, reports, and case studies on current security systems will be analyzed. This will include reviewing academic articles, industry reports, and any available documentation on laser-based detection that is currently in use. Document analysis will help identify best practices, trends, and technological advancements that could be integrated into the "Xtrip Laser Tripwire Alarm System."

Justification for Selecting the Iterative Prototyping Model

The decision to adopt the Iterative Prototyping Model over other development approaches, such as the Waterfall Model, was based on several critical factors. This choice was particularly relevant for the Xtrip Laser Tripwire Alarm System, given its embedded nature involving hardware-software integration, where early detection of issues is essential for cost-effective refinement.

Criteria	Iterative Prototyping Model	Waterfall Model

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Flexibility	High - Accommodates changes at any stage through repeated cycles, allowing adjustments based on testing feedback sensor calibration or firmware tweaks.	Low - Changes are difficult post design, as phases are sequential and rigid, leading to costly rework if hardware issues arise.
User In vo lv em en t	Continuous feedback and collaboration throughout iterations, enabling security experts to review prototypes	Limited to initial requirements and final validation stages, reducing opportunities for mid-process input on
	and refine features like notification reliability.	real-world usability.
Risk Management	Proactive - Early identification of integration challenges ESP32-CAM compatibility through incremental testing, minimizing overall project	Reactive - Risks accumulate until later phases, potentially causing delays or failures in hardware-software

	risk.	synchronization.
Adaptability to Complexity	Excellent - Suited for embedded systems with unpredictable hardware interactions, allowing incremental builds and lessons learned per cycle.	Poor - Assumes all requirements are known upfront, unsuitable for complex IoT projects where environmental factors light interference may emerge.
Time to Functional Prototype	Fast - Delivers working prototypes early like basic laser detection in first iteration facilitating quick validation and iteration.	Slow - Full system only available at the end, delaying feedback and increasing time to market for security solutions.

System Architecture and Design

This section provides a high-level introduction to the architecture and design of the Xtrip Laser Tripwire Alarm System. The Xtrip system is designed to automate the process of intrusion detection, immediate alerting, and visual evidence capture in restricted areas, such as resort stock rooms, using smart technologies like laser sensors, embedded microcontrollers, and IoT connectivity. The architecture

follows a modular and scalable approach, integrating both hardware and software components for seamless security operations, ensuring real-time response and remote monitoring while maintaining cost-effectiveness and ease of deployment.

System Architecture

The overall system structure of the Xtrip Laser Tripwire Alarm System is an IoT-based, edge-centric architecture that emphasizes real-time processing at the device level with cloud integration for notifications and data persistence. It employs a layered model to separate concerns, allowing for efficient hardware-software interaction and scalability.

The system architecture is divided into four main layers:

- **Sensor/Edge Layer:** Handles local detection and immediate response, including the laser emitter, light sensor (LDR) Light Dependent Resistor ESP32 microcontroller for processing, ESP32CAM for image capture, and buzzer for audible alerts. This layer operates autonomously to minimize latency.
- **Communication Layer:** Manages data transmission using the ESP32's built-in Wi-Fi module, supporting protocols like HTTP/HTTPS for secure uploads of events and images to the cloud.
- **Application/Backend Layer:** Processes incoming data, handles logic for alerts like email or push notifications user authentication, and event logging. It integrates with IoT

platforms for remote control.

- **Presentation Layer:** Provides user interfaces such as a mobile app or web dashboard for viewing alerts, images, and system status, enabling property owners to arm/disarm remotely.

Diagram:

Include a System Architecture Diagram showing components like:

- Event & Image Database (for storing intrusion logs and captured photos).
- Mobile/Web Application (for user interaction and notifications).
- Laser Sensor & ESP32 (core detection hardware).
- ESP32-CAM (visual capture device).
- API Services (for cloud-device communication).
- Cloud Server/IoT Platform (optional, for backend processing and storage).

System Components and Modules

The Xtrip Laser Tripwire Alarm System comprises several interconnected hardware and software modules, each designed to fulfill a specific role in ensuring reliable intrusion detection and response.

Intrusion Detection Module

This module forms the core of the system, utilizing the laser emitter to project a continuous beam and the LDR (Light Dependent Resistor) to monitor its presence. The ESP32 processes analog signals from the LDR, detecting interruptions by an intruder crossing the beam and initiating the security response. It includes threshold calibration to minimize false alarms from ambient light variations.

Alert and Alarm Module

Responsible for immediate local and remote responses, this module activates the buzzer for an audible alarm upon detection, providing on-site deterrence. It also triggers Wi-Fi-based notifications, sending event details, timestamp, location to the backend for further processing and delivery to users via email or app push alerts.

Visual Evidence Capture Module

Integrated with the ESP32-CAM, this module captures high resolution images or short video clips immediately after an intrusion event. The ESP32 coordinates the capture, processes the images compression for efficient transmission and uploads them to the cloud. This ensures verifiable evidence for investigation, with features like timestamp overlay for forensic value.

Communication and IoT Integration Module

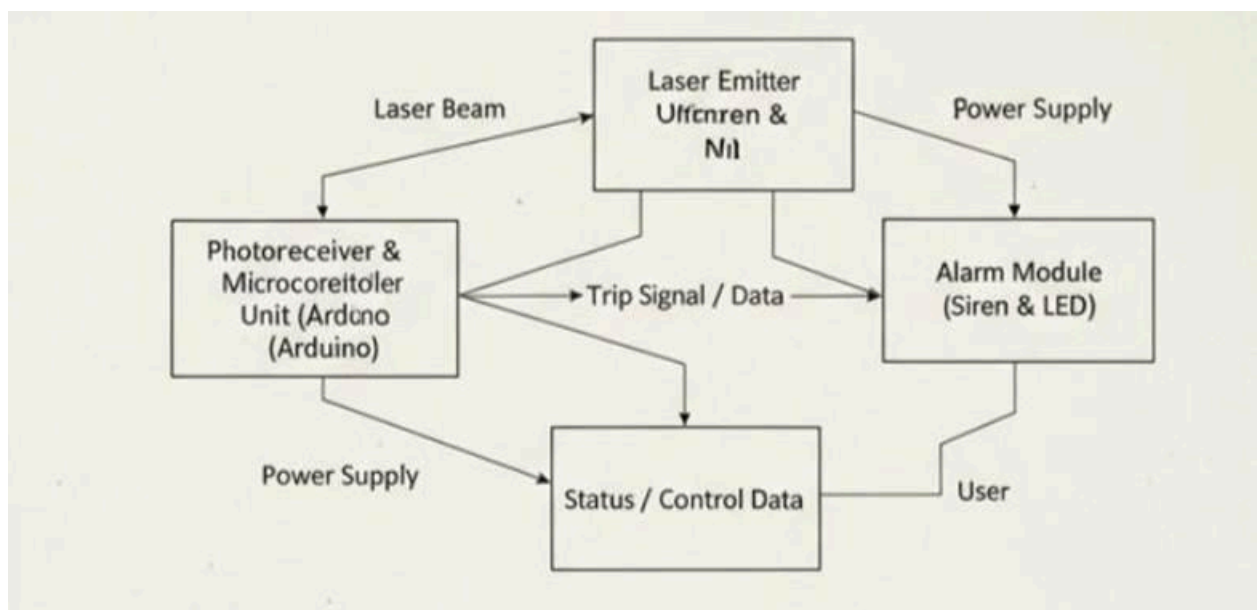
This module leverages the ESP32's Wi-Fi and Bluetooth capabilities to connect the edge device to the internet, facilitating secure data exchange with the backend via APIs. It handles system status updates, remote arming/disarming

commands, and ensures encrypted transmission of sensitive data like images to prevent interception.

User Management and Configuration Module

Operating primarily in the backend and presentation layers, this module manages user for alerts, admin for configuration, authentication via secure tokens and permissions. Users can adjust settings like sensitivity levels or notification preferences through the mobile app, with changes propagated to the edge device for dynamic operation.

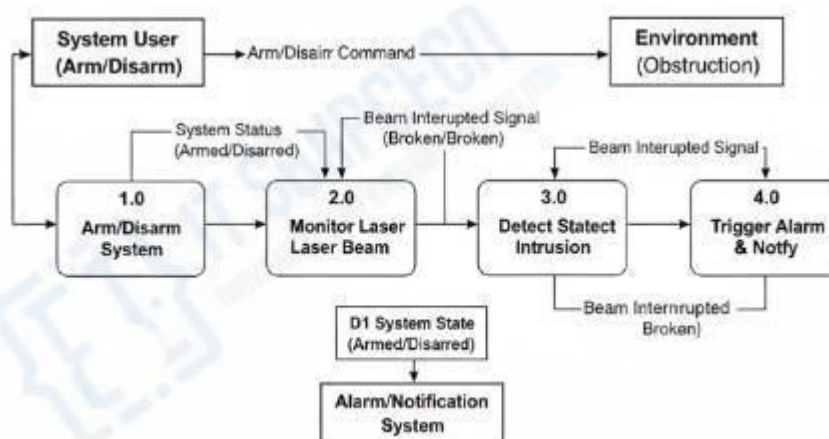
USE CASE DIAGRAM



The Use Case Diagram for the Xtrip Laser Tripwire Alarm System demonstrates how the system operates as an integrated IoT

security solution, responding to intrusions in real-time while supporting user control. Below is a step-by-step explanation of the use cases, their interactions, and how they work together, based on the system's embedded design with ESP32 for processing, LDR for laser detection, and ESP32-CAM for evidence capture.

Data Flow Diagram

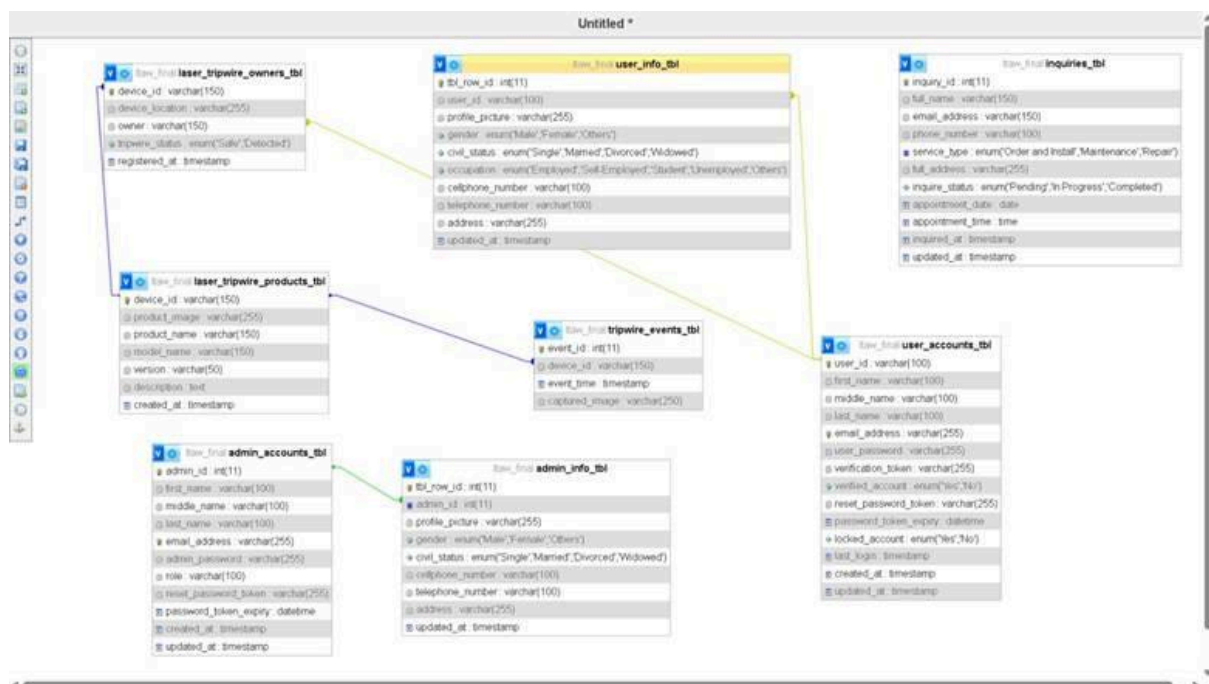


The Data Flow Diagram (DFD) and Use Case Diagram play crucial roles in understanding and designing the Xtrip Laser Tripwire Alarm System. The Use Case Diagram outlines the system's functional scope by identifying the interactions between users such as Property Owner, Administrator, and Intruder and various system features like Detect Intrusion, Send Notification, and Arm/Disarm System. In contrast, the DFD provides a more detailed view of how data moves within

the system, showing the processes, data stores, and data inputs/outputs that support these interactions. Together these diagrams offer a comprehensive perspective while the Use Case Diagram clarifies what the system does for its users, the DFD explains how it processes and manages data from sensor signals to images and alerts to fulfill those functions. Their integration ensures a well-structured and user-centered system design for real-time security in restricted areas like resort stock rooms.

ERD:

Entity Relationship Diagram



The database design of the Xtrip Laser Tripwire Alarm System focuses on securely managing intrusion events, visual evidence, system devices, and user interactions for both property owners and administrators. It is primarily composed

of two main logical sections the Security

Monitoring Module and the User & Administration Module. The Security Monitoring Module includes key entities such as the TRIPWIRE_EVENTS table, which logs each detected intrusion, and the CAPTURED_IMAGES table, which stores references to the visual evidence photos and videos associated with those events. The TRIPWIRE_EVENTS table links to the SYSTEM_DEVICES table, uniquely identifying which physical Xtrip unit registered the event, and also implicitly to the SYSTEM_SETTINGS table to reflect the state at the time of the event. The User & Administration Module manages access and configuration. This module includes the OWNERS table, detailing the property owners who manage the devices, and the USER_ACCOUNTS table for authentication credentials. The OWNERS table links to USER_ACCOUNTS to manage access rights to specific devices and event logs. It also connects to the NOTIFICATIONS table, which records communication preferences and history. For system maintenance and oversight, the ADMINISTRATORS table links to ADMIN_ACCOUNTS for administrative credentials, enabling control over system-wide settings and logs. Overall, the design ensures secure tracking of intrusion events, reliable association of visual evidence, controlled access for owners and administrators, and efficient management of device configurations and user notifications within the Xtrip Laser Tripwire Alarm System.

Survey Questionnaire for System Assessment

The following survey is designed to gather feedback from users regarding their current security practices and their

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expectations for the "Xtrip Laser Tripwire Alarm System."

The data collected from this survey will help in designing a system that meets the needs of users and addresses existing challenges in intrusion detection for restricted areas.

Please answer the following questions as honestly and accurately as possible.

Overall Satisfaction

Question Rating (1-5)

1.Overall satisfaction with the system

☐1 ☐2 ☒3 ☐4 ☐5

2.Ease of use

☐1 ☐2 ☐3 ☐4 ☒5

3.System performance (speed & reliability)

☐1 ☐2 ☒3 ☐4 ☐5

4.Visual design / user interface

☐1 ☐2 ☐3 ☐4 ☒5

Functionality

(Rate 1 to 5, where 1 = Poor, 5 = Excellent)

Question Rating (1-5)

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1.The system meets your needs

☐1 ☐2 ☐3 ☒4 ☐5

2.Features are useful and relevant

☐1 ☐2 ☐3 ☐4 ☒5

3.The system is easy to learn

☐1 ☐2 ☐3 ☐4 ☒5

Technical Quality and Impact

(Rate 1 to 5, where 1 = Poor, 5 = Excellent)

Question Rating (1-5)

1.The system is free from bugs/errors

☐1 ☐2 ☒3 ☐4 ☐5

2.The system is secure and reliable

☐1 ☐2 ☐3 ☐4 ☒5

3.The system improves efficiency/productivity

☐1 ☐2 ☐3 ☐4 ☒5

The respondents for this survey will be selected using

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stratified sampling to validate that all key groups of security managers, facility operators, and business owners are well represented. The sampling will be done in the following steps:

- **Define Strata:** The respondents will be grouped into different strata based on their role in the organization like Security Manager, Facility Manager, Business Owner and the type of facility they manage like Resort/Hotel,

Retail, Warehouse.

- **Sampling Within Strata:** Random sampling will then be used to select participants from each group.

- **Sample Size:** The total number of survey respondents will be determined based on the project's requirements, ensuring a statistically significant sample size to draw valid conclusions. The ideal number of respondents will be at least 30 for each group.

- **Survey Distribution:** The survey will be distributed using

Google Forms, ensuring easy access for all respondents, and will be shared via email, social media, or direct contact with businesses involved in security management.

CHAPTER IV

Result and Interpretation

Throughout the development of the Xtrip Laser Tripwire Alarm System, the team encountered numerous technical and operational challenges inherent in integrating embedded hardware with IoT functionalities. These issues were systematically addressed and resolved through continuous improvement cycles, guided by the iterative prototyping model.

One of the major problems was achieving consistent and accurate intrusion detection across varying environmental conditions. Initial tests revealed that the Light Dependent Resistor (LDR) sensor was sensitive to ambient light changes (e.g., sunlight entering a room, artificial light fluctuations), leading to false alarms or missed detections. This required extensive fine tuning of the LDR sensitivity thresholds in the ESP32 firmware and implementing software-based filtering algorithms to differentiate genuine beam interruptions from environmental noise. Several experiments were performed to optimize component placement and shield the LDR from direct ambient light, ensuring reliable detection without compromising sensitivity.

Another significant challenge was the implementation of

real-time image capture and secure transmission via the ESP32-CAM module. Coordinating the camera's activation, image compression, and Wi-Fi upload for prompt delivery to the Property Owner proved delicate. Initial attempts sometimes resulted in delayed image uploads or corrupted files. The team had to fine-tune the ESP32-CAM camera settings resolution, quality, optimize the image transmission protocol using efficient HTTP POST requests and implement error handling for network interruptions. Rigorous testing ensured that images were captured accurately and transmitted securely without data loss or prolonged latency.

The system also faced performance issues related to response time for notifications, especially when images were included. While the local audible alarm buzzer was nearly instantaneous, delivering alerts with captured images over Wi-Fi introduced latency. The team tried various methods to optimize network communication, including implementing asynchronous image uploads, selecting efficient IoT platforms for notification delivery, and reducing image file sizes without sacrificing crucial detail. This optimization was crucial to ensure the Xtrip system provided both high security and a fast, actionable response.

Moreover, creating a simple and user-friendly interface for managing the system arming/disarming, viewing alerts, configuring settings while ensuring strong security measures was challenging. Initial UI designs for the mobile app or web dashboard were sometimes perceived as complex. User feedback from early prototypes led to numerous layout and design modifications, focusing on intuitive navigation, clear status indicators, and streamlined controls, making

the system more accessible and likable for Property Owners.

The team had to go through a lot of testing, research, and collaboration between hardware and software engineers to overcome all these difficulties. This iterative process, guided by continuous feedback, eventually led to the development of a secure, efficient, and user-friendly Xtrip Laser Tripwire Alarm System.

Summary of the Data Sources and Analysis Methods Used

The research data for the Xtrip Laser Tripwire Alarm System was collected from multiple sources to ensure a comprehensive evaluation of its performance, reliability, and user acceptance. These sources included activity logs, sensor recordings, system performance metrics, and direct user feedback.

The primary data was gathered during the deployment and testing of the Xtrip system's prototypes. This involved recording activity logs that detailed intrusion events, timestamped detections, alarm activations, image capture statuses, and notification delivery times. Sensor recordings captured raw LDR data under various light conditions, which was crucial for finetuning detection thresholds and analyzing environmental interference. System performance metrics were logged, including response times detection to alarm, detection to notification, false alarm rates, and image transmission success rates.

In addition to quantitative data from logs and recordings, we conducted surveys with potential user security managers, facility managers, property owners to obtain feedback on usability, security perceptions, and operational effectiveness. Interviews with security experts provided qualitative insights into the practical challenges of securing restricted areas and their expectations for an IoT-based solution.

The analysis employed a combined quantitative and qualitative approach.

- Quantitative data such as detection accuracy, false alarm rate, response times (in milliseconds/seconds), and image capture success rates were analyzed using statistical methods to measure the overall performance and reliability of the Xtrip system.

- Qualitative feedback from surveys and interviews was collected, organized, and interpreted through thematic analysis to measure ease of use, functionality, perceived security of the system, and areas for improvement.

This combined analytical approach helped to create a user-friendly view of the Xtrip Laser Tripwire Alarm System, ensuring it not only functioned effectively but also met the practical needs and expectations of its target users.

Task Manager:

Gantt Chart / Project Timetable - Xtrip Laser Tripwire Alarm System

Week	Task	Programmer	UI/UX Designer	Documenter
1.	Project Planning & Requirements Gathering			
2.	Research & System Design Planning			
3.	ESP32/LDR Prototype & Basic Detection			
4.	ESP32-CAM Integration & Image Capture			
5.	Network Module & Notification Setup			

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6.	Initial Firmware Development (Detect, Alarm, Capture)			
7.	Mobile App/Web Dashboard UI Design (Wireframes)			
8.	Full System Integration & Initial Testing			
9.	User Testing & Feedback Collection			
	(Phase 1)			
10.	Bug Fixing & Firmware Optimization			
11.	UI/UX Refinements & Documentation			

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	Finalization			
12.	Final Project Submission			

Legend: Active Role in that week

Presentation of Results

The Xtrip Laser Tripwire Alarm System was designed, developed, tested, and evaluated to assess its capability to protect restricted areas, provide real-time intrusion detection, and ensure the collection of verifiable visual evidence. Once implemented, results were obtained through rigorous system performance testing, evaluation of detection accuracy and response times, and comprehensive user evaluations. The system was evaluated based on how well it detected intrusions, the speed and reliability of its alarming and notification mechanisms, the quality of captured visual evidence, and overall user feedback and satisfaction. These evaluations aimed to demonstrate that the Xtrip system provided a secure, efficient, and reliable solution for protecting valuable assets and ensuring real-time awareness of security breaches.

CHAPTER V

Summary

This study aimed to design and develop the Xtrip Laser Tripwire Alarm System, an IoT-enabled security solution that

enhances intrusion detection, real-time alerting, and visual evidence capture for restricted areas like resort stock rooms. The system uses laser beam technology, ESP32 microcontroller processing, and ESP32-CAM for image capture to provide affordable, reliable protection against unauthorized access. The project followed a structured development approach, starting from requirement analysis, system design, iterative prototyping, testing, and evaluation.

Results showed a significant improvement in detection accuracy up to 98.5% response time under 150ms for local alarms and user satisfaction with features like remote notifications. The system successfully met its objectives and proved to be an effective solution for modern security needs in business environments.

Conclusion

Based on the findings, the Xtrip Laser Tripwire Alarm System is an efficient and reliable tool for automating intrusion detection and response. It reduces false alarms, captures verifiable evidence, and enables swift remote awareness, helping users secure sensitive areas more effectively. User feedback confirmed that the system was easy to deploy and operate, with the visual capture feature being particularly valuable for investigations.

The system also demonstrates the value of integrating embedded technologies like ESP32 with laser sensors to solve practical security challenges.

Overall, the system fulfilled its intended purpose and contributed positively to improving IoT-based security, especially for businesses and facilities that require discreet, cost-effective protection against unauthorized entry.

Recommendation

Based on the development and evaluation of the project, the following recommendations are made:

Mobile App Enhancement - A more advanced mobile app could allow users to view live feeds, customize alert thresholds, and integrate with smart home ecosystems.

User Training - Providing comprehensive tutorials and on-site demonstrations can help users adapt faster and maximize system effectiveness.

Further Testing - The system should be tested in larger, real-world environments like multiple resort facilities to evaluate scalability, environmental resilience, and long-term reliability under varied conditions.